COM4290/6131 Tutorial 1

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1. Leaves fall from a tree to the ground as a Poisson process at a rate of one every minute.

   (a) You arrive at 4:00 pm. During the next ten minutes, what is the probability that you see twelve leaves fall?

   (b) What is the probability that you see n leaves fall where n goes from 0 to 20? Use MATLAB! Plot the results on a graph.

   (c) Where does the curve peak? Compare the peak with the expected value, λt.

2. X takes on four different values with differing probabilities, as shown in the following table.

X	1	2	3	4
$P(X)$	0.05	0.2	0.3	0.45

   (a) What is the mean of X?

   (b) What is the standard deviation of X?

   (c) What is the expectation of $\ln X$?

3. Show that

$$\left\langle n^2 \right\rangle = \lambda t e^{-\lambda t} \sum_{n=1}^{\infty} \frac{n(\lambda t)^{(n-1)}}{(n-1)!}, \tag{1}$$

   where $\left\langle n^2 \right\rangle$ represents the expectation of n^2 under a Poisson distribution. Then by making the substitution $m = n - 1$ show that

$$\left\langle n^2 \right\rangle = (\lambda t)^2 + \lambda t. \tag{2}$$

   What, therefore, is the variance of a Poisson distribution?

   Hint: There are similarities with the proof of the mean under a Poisson distribution we went through in the lecture.

4. **This question will require you to use MATLAB.** Use the function `randn` to take a set of N samples, $\{x_n\}_{n=1}^{N}$ from a Gaussian with mean 0 and variance 1. For values of N given in the table below, compute the sample mean and the sample variance.

N	10	50	100	200	500	1000
$\frac{1}{N}\sum_{n=1}^{N} x_n$						
$\frac{1}{N}\sum_{n=1}^{N} x_n^2$						

(a) What is the true *mean* of the distribution you've sampled from?

(b) For each point, plot the squared difference between the *sample mean* and the true *distribution mean* against the values of N given. What sort of trend do they give? Is that trend consistent? When, typically, are the sample based estimates of the mean worst?

(c) What is the true *second moment* of the distribution you've sampled form?

(d) For each point, plot the squared difference between the *sample based estimate* of the second moment and the *true* second moment of the distribution. Is the trend similar to that of the sample based mean? Are the estimates of the second moment typically better or worse? Why do you think this is the case?

Hint: You may want to summarise the quality of your results by repeating the experiment multiple times and looking at the mean and standard deviations. You might find the MATLAB functions `plot` *and errorbar useful when doing this.*

5. Compute the mean and variance of the exponential distribution,

$$p(x) = \lambda \exp(-\lambda x), \quad \text{where } 0 > x > \infty.$$

6. Download the code for sampling from an exponential distribution from `http://www.dcs.shef.ac.uk/~neil/campus_only/com6131/`. Repeat the experiments in Quesion 4 using an exponential distribution with a rate parameter of $\lambda = 3$.

N	10	50	100	200	500	1000
$\frac{1}{N}\sum_{n=1}^{N} x_n$						
$\frac{1}{N}\sum_{n=1}^{N} x_n^2$						